Fiscal Year:	FY 2021	**Task Last Updated:**	FY 09/23/2020
PI Name:	Gilroy, Simon Ph.D.		
Project Title:	BRIC: Exploring Spaceflight-Linked Changes in Plant Defense Capabilities		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology (3) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential (2) Bioregenerative Life Support		
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Comments:	NOTE: PI formerly at Pennsylvania State University; moved to University of Wisconsin-Madison in 2007 (Info received 7/2009)		
Project Type:	Flight	**Solicitation / Funding Source:**	2016-17 Space Biology (ROSBio) NNH16ZTT001N-FG. App G: Flight and Ground Space Biology Research
Start Date:	10/26/2018	**End Date:**	10/25/2021
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	15	**Monitoring Center:**	NASA KSC
Contact Monitor:	Massa, Gioia	**Contact Phone:**	321-861-2938
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Flight Program:	ISS		
Flight Assignment:	NOTE: End date changed to 10/25/2021 per G. Massa/KSC; forthcoming in NSSC (Ed., 9/24/20)		
Key Personnel Changes/Previous PI:	September 2020 report: None		
COI Name (Institution):	Swanson, Sarah Ph.D. (University of Wisconsin, Madison)		
Grant/Contract No.:	80NSSC19K0126		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Analysis of the plant biology transcriptomics and proteomics data in the GeneLab data system reveals that a common feature of spaceflight is alteration in plant defense-related transcripts and proteins in space. Anecdotally, plants have also been seen to suffer from pathogen attack during spaceflight missions. Taken together, these observations suggest a hypothesis that plant defense responses may be compromised by spaceflight. Such a disruption of defense would have important implications for the management of plants grown as part of an astronaut life support system. This is especially relevant given the large microbial community on board the International Space Station (ISS) and the possibility that spaceflight triggers enhanced microbial pathogenicity. However, the critical unanswered question is whether the altered defense gene expression seen in spaceflight truly plays out as altered susceptibility to pathogens. Use of the Biological Research in Canisters-Light Emitting Diodes (BRIC-LED) hardware growing Arabidopsis seedlings allows direct testing of this idea. Specifically, the project will: (1) Capitalize on the unique capabilities of the BRIC-LED hardware to analyze the transcriptome and cell wall structure responses of Arabidopsis seedlings grown in spaceflight. Plants will be grown in the hardware and then challenged with the bacterial elicitor peptide flg22. Comparison of subsequent on-orbit responses in wild-type and mutants in the defense regulators AtRBOHD and CML24 will then be made. (2) Use comparisons to ground-based control experiments and clinostat and hypoxia treatment to ask whether such changes are likely linked to the microgravity or hypoxic elements of spaceflight and whether these changes may affect the ability of the plant to defend itself against pathogen challenge. The BRIC-LED/Petri Dish Fixation Unit (PDFU) hardware is well-suited to this experimental approach. Thus, the two-stage PDFU hardware will allow both stimulation (injection of flg22 elicitor onto the seedlings) and subsequent fixation of samples (with RNAlater). As the experiment would use flg22 peptide to elicit defense response, potential problems with flying live pathogen will be circumvented. The experimental design will capitalize on the minimal crew time needed for a BRIC-based experiment and the unique capabilities of the BRIC hardware to treat plants in situ on orbit. The results from this experiment will allow direct testing of the critical question of whether the ability of plants to mount a defense response is compromised during spaceflight.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is addressing how pathogen sensing and response operates in plants, targeting the Ca2+-dependent and reactive oxygen signaling machinery. The results from this work are not only providing insight into how pathogen signals impact plant growth and defense responses in space but also are contributing to our understanding of similar responses on Earth. Pathogen attack is a major limit on crop productivity with estimates of global losses in the range of 20% across multiple major crops such as wheat, maize, rice, soybean, and potato. The pathogen sensing and response systems triggered by the molecular signals of bacterial attack in space are also those employed by plants challenged by pathogens on Earth. Thus, this work aimed at analysis of how reactive oxygen species and Ca2+ signaling is involved in the defense response systems induced during spaceflight should also help our understanding for these same systems in this general plant defense response system on Earth. These results will provide information to highlight novel molecular targets for potential manipulation to tailor plant responses to the wide spectrum of pathogens that attack plants.
Task Progress:	In the previous granting period, protocols and procedures had been optimized for growing the Arabidopsis plants that are the focus of this investigation in the BRIC-LED hardware. Therefore, Science Verification Testing and Experiment Verification Testing (EVT) were performed September 2019 and January 2020 as the next steps in preparation for flight. These tests are designed to optimize protocols and procedures and also to spot and solve potential, unexpected issues that emerge as the experiment is performed exactly as it would be on the International Space Station. The Science Verification Testing was successful with all outcomes (germination, plant growth, and RNA isolation) being in the excellent/good range from the predefined mission success criteria. The Experiment Verification Testing was then performed in January 2020 using the protocols refined during the Science Verification test. This testing provided critical feedback revealing the potential for a cryptic failure point for the internal lighting system that could be eliminated with a small change in circuitry design. A new EVT is planned later in 2020 to test these enhanced units. Ground-based analysis of pathogen response: The focus of BRIC-LED 002 is to understand how plant defense is affected by spaceflight. Any alterations in defense signaling would likely be related to one or more aspects of the flight environment such as microgravity, development of hypoxia, and/or increased radiation. BRIC-LED 002 is also comparing responses in wild type to two sets of mutants in the genes for the proteins Calmodulin-like 24 and AtRBOHD. Both of these genes are linked to plant defense signaling through cellular Ca2+ and reactive oxygen-related signaling systems. We have therefore been using a battery of bioinformatic, physiological, and gene expression assays in ground-based analyses to ask how these genes might play roles in stress signaling related to the expected stresses in spaceflight and how these might then affect pathogen defense. We have been analyzing potential relationships to hypoxia and flooding-related stresses as these might be encountered in the microgravity environment of spaceflight. These ground-based studies have also been comparing wild-type plants to the knockout mutants in CML24 and RBOHD that will be flown on BRIC-LED 002. From these studies, we have found that both genes are involved in the response to long-term flooding challenge. As both CML24 and RBOHD are related to Ca2+-dependent signaling, we have also been working on defining how the cellular Ca2+ response network might be integrated into their defense and flooding signaling systems. The dynamics of these cellular Ca2+ changes are regulated by the Ca2+-calmodulin-regulated ATPases. We have found that two of these ATPase Ca2+ pumps, ACA4 and ACA11, are required for Ca2+ signaling and subsequent pathogen signaling. These pumps are found on the internal vacuolar membrane of the cell and unexpectedly, we were able to rescue the Ca2+ signaling and the defense abilities of the knockout mutant in both by redirecting a pump from the plasma membrane to the vacuole, suggesting an unexpected flexibility in the pathogen Ca2+ signaling system and indicating that the vacuole may be a major site of pathogen signaling processing. We have also continued to develop bioinformatics-based tools to be able to compare ground-based and flight samples in the BRIC hardware to infer response systems linked to reactive oxygen species-ROS (RBOHD) and Ca2+ (CML24) that are expected in the BRIC-LED flight samples. The bioinformatics engine developed to allow these kinds of analyses, the TOAST database, was published in early 2020 in Frontiers in Plant Science (See Bibliography section). Presentations, Outreach and Education: During 2019-2020, we have presented this research at the annual meeting of the American Society for Gravitational and Space Research and the ISS R&D conference. We have also used our space

	research as a base for our outreach efforts, where we have used it in projects for under-represented groups in STEM (Science, Technology, Engineering, and Mathematics), such as the PEOPLE program, in presentations for high school students and undergraduates (e.g., University of Wisconsin-Madison's BioHouse), to K-12 teachers via courses for the BioPharmaceutical Technology Center's Summer training program and to a program for California high school teachers, and internationally, e.g., to school students in Mumbai and Calcutta, India. We continue to work closely with Madison West High School's Rocketry Society. These students have designed, built, and flown Arabidopsis experiments on their rocket flights and have analyzed hypergravity profiles taken from their rocket launches, then using QPCR to define how these stresses alter defense signaling. This is a continuing program where we are developing a pipeline of talented and engaged students who are going on to College in STEM areas. We continue to maintain a "Collaboratory" where biology and engineering undergraduates and high school students come together to develop high throughput phenotyping equipment and other hardware relevant to our spaceflight-related work. Over 2019-2020 we have mentored approximately 10 students as part of this program, working on various plant molecular and engineering projects related to space biology and BRIC-LED-related research, with a team developing mechanical stimulation systems for plants (based on vibrations or wind/rain simulators) to define mechano-response to these effects. We also work closely with the Insight Wisconsin program for undergraduate inventors. Our current team here is designing a clinostat for high throughput screening for spaceflight-related phenotypes.
Bibliography Type:	Description: (Last Updated: 02/02/2026)
Articles in Peer-reviewed Journals	Hilleary R, Paez-Valencia J, Vens C, Toyota M, Palmgren M, Gilroy S. "Tonoplast-localized Ca 2+ pumps regulate Ca 2+ signals during pattern-triggered immunity in Arabidopsis thaliana." Proc Natl Acad Sci U S A. 2020 Aug 4;117(31):18849-57. https://doi.org/10.1073/pnas.2004183117 ; PMID: 32690691; PMCID: PMC7414185 , Aug-2020
Articles in Peer-reviewed Journals	Barker R, Lombardino J, Rasmussen K, Gilroy S. "Test of Arabidopsis Space Transcriptome: A discovery environment to explore multiple plant biology spaceflight experiments." Front Plant Sci. 2020 Mar 4;11:147. https://doi.org/10.3389/fpls.2020.00147 , PMID: 32265943; PMCID: PMC7076552 , Mar-2020
Articles in Peer-reviewed Journals	Neelam S, Richardson B, Barker R, Udave C, Gilroy S, Cameron MJ, Levine HG, Zhang Y. "Changes in nuclear shape and gene expression in response to simulated microgravity are LINC complex-dependent." Int J Mol Sci. 2020 Sep 15;21(18):E6762. https://doi.org/10.3390/ijms21186762 ; PMID: 32942630 , Sep-2020